MILITARY SPECIFICATION

WAVEGUIDES, ELLIPTICAL

*GENERAL SPECIFICATION FOR*

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for flexible and semirigid waveguides with elliptical inside configuration (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| L-P-390 | - | Plastic, Molding and Extension Material, Polyethylene and Copolymers (Low, Medium, and High Density). |
| QQ-S-781 | - | Strapping, Steel, Flat and Seals. |
| PPP-B-566 | - | Boxes, Folding, Paperboard. |
| PPP-B-585 | - | Boxes, Wood, Wirebound. |
| PPP-B-601 | - | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | - | Boxes, Shipping, Fiberboard. |
| PPP-B-676 | - | Boxes, Setup. |
| PPP-T-60 | - | Tape: Packaging, Waterproof. |
| PPP-T-66 | - | Tape, Packaging, Vinyl Plastic Film. |
| PPP-T-76 | - | Tape, Packaging, Paper, (for Carton Sealing). |
| PPP-T-495 | - | Tube, Mailing, and Filing. |

Beneficial comments (recommendations, additions, and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 5043, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document or by letter.



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| | | |
|---|---|---|
| MIL-P-116 | - | Preservation-Packaging, Methods of. |
| MIL-B-131 | - | Barrier Materials, Watervaporproof, Flexible, Heat-Sealable. |
| MIL-C-5541 | - | Chemical Conversion Coatings on Aluminum and Aluminum Alloys. |
| MIL-C-12000 | - | Cable, Cord, and Wire, Electric; Packaging of. |
| MIL-F-22191 | - | Barrier Materials, Transparent, Flexible, Heat Sealable. |
| MIL-W-28839/1 | - | Waveguides, Elliptical, Semirigid. |

STANDARDS

FEDERAL

Federal STD 356 - Commercial Packaging of Supplies and Equipment.

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - | Marking for Shipment and Storage. |
| MIL-STD-202 | - | Test Methods for Electronics and Electrical Component Parts. |
| MIL-STD-1186 | - | Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods. |
| MIL-STD-1285 | - | Marking of Electrical and Electronic Parts. |

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | | |
|---|---|---|
| B241 | - | Aluminum-Alloy Extruded Bars, Rods and Tubes. |
| B152 | - | Copper Sheet, Strip, Plate, and Rolled Bar. |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), the contractor shall furnish a sample unit for first article inspection and approval (see 4.5 and 6.3).

3.3 Materials. The material shall be as specified (see 3.1). Acceptance or approval of any constituent material shall not be construed as a guaranty of the finished product.

3.3.1 Aluminum alloy. Aluminum alloys 1060, 1100, 6061, and 6063 shall conform to the chemical composition limits of ASTM B241. The minimum resistivity shall be as follows:

a. 1060, ____ micro ohms/centimeter.
b. 1100, 2.90 micro ohms/centimeter.
c. 6061, 4.0 micro ohms/centimeter.
d. 6063, 3.4 micro ohms/centimeter.



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3.3.2 Copper. Copper type OF shall conform to ASTM B152.

3.3.3 Jacket. Waveguide jacket material shall conform to L-P-390, type III, class H.

3.4 Design and construction. Waveguides shall be of the design, construction, and physical dimensions specified (see 3.1).

3.4.1 Finish. The interior and exterior (except for jacketed waveguides) surfaces of the waveguide shall be bright, smooth, dry, and free of scales or oxide. All aluminum alloy waveguides shall be chemically treated in accordance with MIL-C-5541.

3.4.2 Length. The length of the finished waveguide shall be as specified (see 3.1). The waveguide length tolerance shall be in accordance with table I.

TABLE I. Length tolerances.

| Waveguide length | Tolerance |
|---|---|
| (feet) | |
| 0-25 | $^{-0}_{+1}$ inch |
| 25 or greater | ±2 percent |

3.5 Voltage standing wave ratio (VSWR). When waveguides are tested as specified in 4.7.2, the VSWR shall not exceed the value specified (see 3.1).

3.6 Attenuation. When waveguides are tested as specified in 4.7.3, the attenuation shall not exceed the value specified (see 3.1).

3.7 Pressurization (flexible type only). When waveguides are tested as specified in 4.7.4, there shall be no leakage, no distortion of aperture due to pressure, and the pressure shall not have dropped more than 2 $lbf/ft^2$. After the test, the VSWR shall be as specified in 3.5.

3.8 Bending radius (flexible type only). When waveguides are tested as specified in 4.7.5, the bend shall be smooth and continuous without distortion of the cross-sectional aperture. After this test, the VSWR shall be as specified in 3.5.

3.9 Axial twist. When waveguides are tested as specified in 4.7.6, the twist shall be smooth and continuous without distortion of the cross-sectional aperture. After this test, the VSWR shall be as specified in 3.5.

3.10 Vibration, high frequency (flexible type only). When waveguides are tested as specified in 4.7.7, there shall be no mechanical damage to the jacket or the metal structure, as applicable. After this test, the VSWR and attenuation shall be as specified in 3.5 and 3.6, respectively.

3.11 Shock (specified pulse) (flexible type only). When waveguides are tested as specified in 4.7.8, there shall be no mechanical damage to the jacket or the metal structure, as applicable. After this test, the VSWR and attenuation shall be as specified in 3.5 and 3.6, respectively.

3.12 Moisture resistance (flexible type only). When waveguides are tested as specified in 4.7.9, there shall be no evidence of cracking or separation of the jacket, as applicable. After this test, the VSWR and attenuation shall be as specified in 3.5 and 3.6, respectively.



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3.13 Thermal shock (flexible type only). When waveguides are tested as specified in 4.7.10, there shall be no evidence of damage to the jacket or metal structure. After this test, the VSWR and attenuation shall be as specified in 3.5 and 3.6, respectively.

3.14 Power handling capability (when specified, see 3.1 and 6.2). When waveguides are tested as specified in 4.7.11, there shall be no evidence of mechanical damage to the metal structure or the jacket, as applicable. After this test, the VSWR shall be as specified in 3.5.

3.15 Surface roughness. When waveguides are tested as specified in 4.7.12, the interior surface roughness shall not exceed the value specified (see 3.1).

3.16 Marking. The waveguide shall be marked in accordance with method I of MIL-STD-1285, with the part number and the manufacturer's source code. The manufacturer's name or trademark may also be marked on the waveguide, provided such letters are not expressly forbidden in the contract. The marking shall be applied with a permanent ink or paint along one side of the waveguide. Not more than 12-inch intervals shall be permitted from the end of one marking to the beginning of the next marking.

3.17 Workmanship. The waveguide shall be free from defects of a nature that may interfere with normal military applications. The waveguide shall be uniform in composition, free from internal and external mechanical imperfections, and shall have a clean, bright appearance in accordance with good commercial practice. The interior surface of the waveguide shall be free from burrs, die marks, dirt, grease, scales, and splinters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Materials inspection (see 4.4).
b. First article inspection (see 4.5).
c. Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table II, used in fabricating the waveguide, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE II. Materials inspection.

| Material | Requirement paragraph | Applicable specification |
|---|---|---|
| Aluminum alloy | 3.3.1 | ASTM B241 |
| Copper | 3.3.2 | ASTM B152 |
| Jacket | 3.3.3 | L-P-390 |



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4.5 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on a sample unit which has been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.5.1 Sample size. One waveguide shall be subjected to first article inspection.

4.5.2 Inspection routine. The sample shall be subjected to the inspections specified in table III, in the order shown.

TABLE III. First article inspection.

| Inspection | Requirement paragraph | Test method paragraph |
|---|---|---|
| Visual and mechanical inspection | 3.3 thru 3.4.2, 3.16, and 3.17 | 4.7.1 |
| VSWR | 3.5 | 4.7.2 |
| Attenuation | 3.6 | 4.7.3 |
| Pressurization 1/ | 3.7 | 4.7.4 |
| Bending radius | 3.8 | 4.7.5 |
| Axial twist | 3.9 | 4.7.6 |
| Vibration, high frequency 1/ | 3.10 | 4.7.7 |
| Shock (specified pulse) 1/ | 3.11 | 4.7.8 |
| Moisture resistance 1/ | 3.12 | 4.7.9 |
| Thermal shock 1/ | 3.13 | 4.7.10 |
| Power handling capability 2/ | 3.14 | 4.7.11 |
| Surface roughness | 3.15 | 4.7.12 |

1/ Flexible type only.
2/ When specified (see 3.1 and 6.2).

4.5.3 Failures. One failure shall be cause for refusal to grant first article approval.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. An inspection of product for delivery shall consist of groups A and B inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all the waveguides of the same part number, produced under essentially the same conditions, and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table IV, in the order shown.

TABLE IV. Group A inspection.

| Inspection | Requirement paragraph | Test method paragraph | AQL (percent defective) | |
|---|---|---|---|---|
| | | | Major | Minor |
| Visual and mechanical inspection | 3.3 thru 3.4.2, 3.16, and 3.17 | 4.7.1 | 1.0 | 4.0 |
| VSWR | 3.5 | 4.7.2 | | |
| Attenuation | 3.6 | 4.7.3 | | |
| Pressurization 1/ | 3.7 | 4.7.4 | 1.0 | 4.0 |

1/ Flexible type only.



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4.6.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table IV. Major and minor defects shall be as defined in MIL-STD-105.

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table V, in the order shown. The sample for group B inspection shall be selected from inspection lots that have passed the group A inspection.

TABLE V. Group B inspection.

| Inspection | Requirement paragraph | Test method paragraph |
|---|---|---|
| Bending radius | 3.8 | 4.7.5 |
| Axial twist | 3.9 | 4.7.6 |
| Vibration 1/ | 3.10 | 4.7.7 |
| Shock (specified pulse) 1/ | 3.11 | 4.7.8 |
| Moisture resistance 1/ | 3.12 | 4.7.9 |
| Thermal shock 1/ | 3.13 | 4.7.10 |
| Power handling capability 2/ | 3.14 | 4.7.11 |

1/ Flexible type only.
2/ When specified (see 3.1 and 6.2).

4.6.1.3.1 Sampling plan. One length of the production lot shall be tested. If the size of the production lot is greater than 25 pieces, one additional length shall be tested per each 25 additional pieces in the lot.

4.6.1.3.2 Failures. If one or more sample units fail to pass group B inspection, the lot shall be considered to have failed.

4.6.1.3.3 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract.

4.6.2 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

4.7 Methods of inspection.

4.7.1 Visual and mechanical inspection. Waveguides shall be examined to verify that the materials, design, construction, physical dimensions, finish, length, marking and workmanship are in accordance with the applicable requirements (see 3.3 thru 3.4.2, 3.16 and 3.17).

4.7.2 Voltage standing wave ratio (VSWR, see 3.5). The VSWR of the waveguide shall be measured over the specified frequency range (see 3.1), using the test set up of figure 1. The test equipment shall be capable of providing a continuous measurement of VSWR over the required frequency range. A means shall be provided for producing a permanent record of the waveguides' VSWR versus frequency. The permanent record



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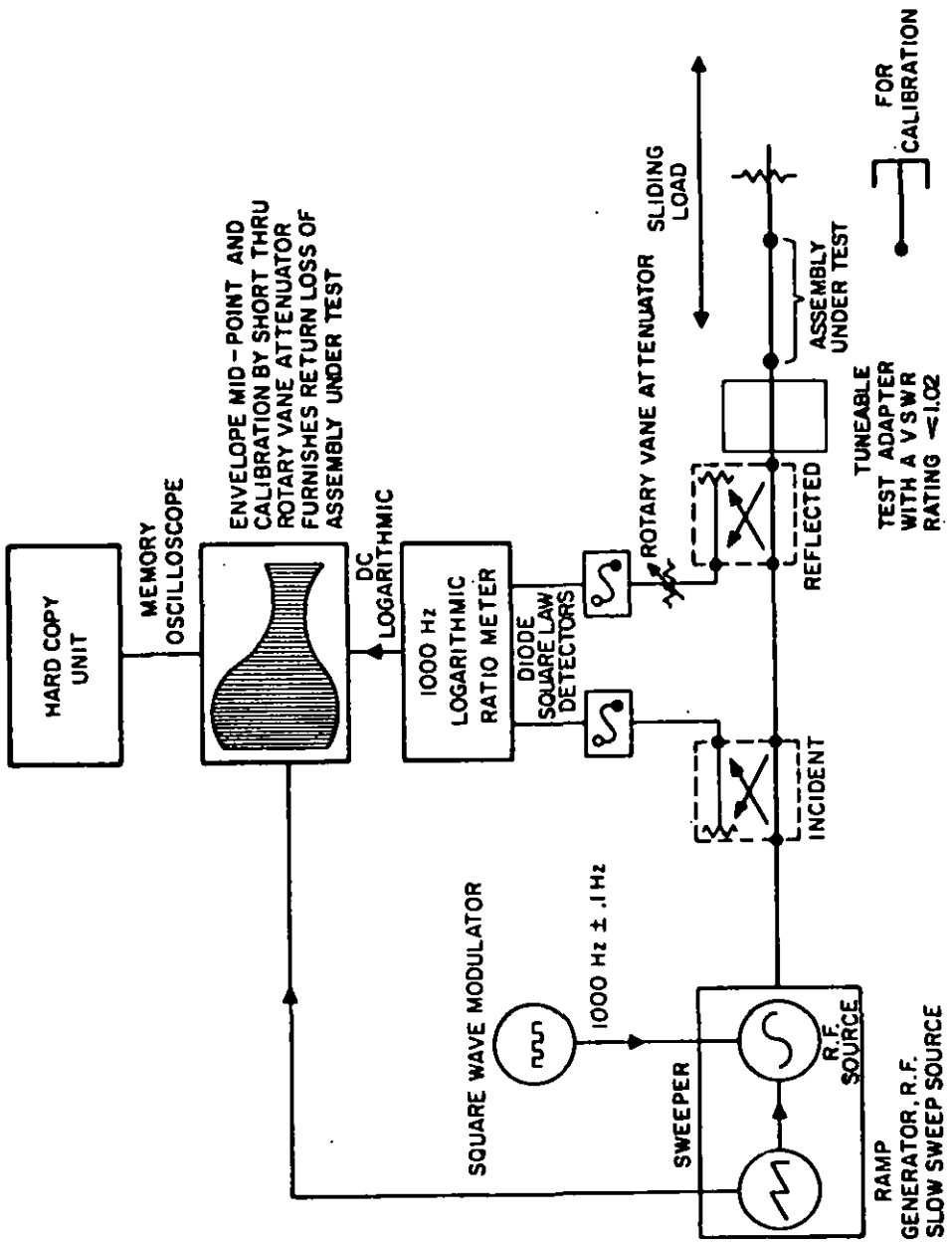



FIGURE 1. VSWR test setup.



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shall indicate the worst case VSWR numerically and shall provide the calculation used to obtain the calculated VSWR. The measurement system and permanent record shall provide a minimum accuracy of .03 in VSWR other test method of equal accuracy may be used.

4.7.3 Attenuation (see 3.6). The attenuation of the waveguide shall be measured over the specified frequency range (see 3.1), using the test set-up of figure 2. The test equipment shall be capable of providing a continuous measurement of attenuation over the required frequency range. A means shall be provided to produce a permanent record of the waveguide's attenuation versus frequency. The measurement system and permanent record shall provide a minimum accuracy of .01 dB for the frequency range of 2.60 thru 26.5 GHZ and .02dB for frequency ranges of .32 through 2.60 GHZ and 26.5 through 90 GHZ. The zero trace is formed by a measurement of the adapters back-to-back (the normalizer could then subtract this attenuation.. Other test methods of equal accuracy may be used.

4.7.4 Pressurization (see 3.7). The waveguide shall be subjected to the internal air pressure in pound-force per square foot ($lb_f/ft^2$) specified (see 3.1). The pressurization supply shall be disconnected and pressure maintained in the waveguide for 4 hours. At the end of the 4-hours period, the waveguide pressure shall not have dropped more than 2 $lb_f/ft^2$. The VSWR shall be measured while under pressure as specified in 4.7.2.

4.7.5 Bending radius (see 3.8). One end of the waveguide shall be fixed to a stationary platform with the waveguide supported in a horizontal position. The free end of the waveguide shall then be bent around a circular mandrel with the specified radius (see 3.1), to at least a 180-degree bend. A representative sample of the same part number as the waveguide under test may be used for this test. The length of the representative sample shall be sufficient to complete the 180-degree bend. After completion of the bend, the VSWR shall be measured as specified in 4.7.2.

4.7.6 Axial twist (see 3.9). One end of the waveguide shall be fastened to a stationary platform with the waveguide supported in a horizontal position. To prevent the waveguide from bending, a rigid shaft shall be extended through the waveguide and fastened to the stationary platform. The free end of the waveguide shall be twisted through a 90-degree angle. The length of the twisted section shall be as specified (see 3.1). A representative sample of the same part number as the waveguide under test may be used for the axial twist test. The length of the representative sample shall be sufficient to complete the 90-degree twist. After completion of the twist, the VSWR shall be measured as specified in 4.7.2.

4.7.7 Vibration, high frequency (see 3.10). Waveguides shall be tested in accordance with MIL-STD-202, method 204. The following details shall apply:

a. Mounting - Unless otherwise specified (see 3.1), rigidly mounted to an appropriate nonresonant mounting table.
b. Electrical load - Not applicable.
c. Test condition letter - A.
d. Resonance - Not applicable.
e. Measurements after vibration - VSWR and attenuation shall be measured as specified in 4.7.2 and 4.7.3, respectively.

4.7.8 Shock (specified pulse (see 3.11). Waveguides shall be tested in accordance with MIL-STD-202, method 213. The following details shall apply:

a. Mounting - Unless otherwise specified (see 3.1), rigidly mounted to test platform.
b. Test condition letter - G.
c. Measurements after shock - VSWR and attenuation shall be measured as specified in 4.7.2 and 4.7.3, respectively.

4.7.9 Moisture resistance (see 3.12). Waveguides shall be tested in accordance with MIL-STD-202, method 106. The following details shall apply:

a. Initial measurements - Not applicable.



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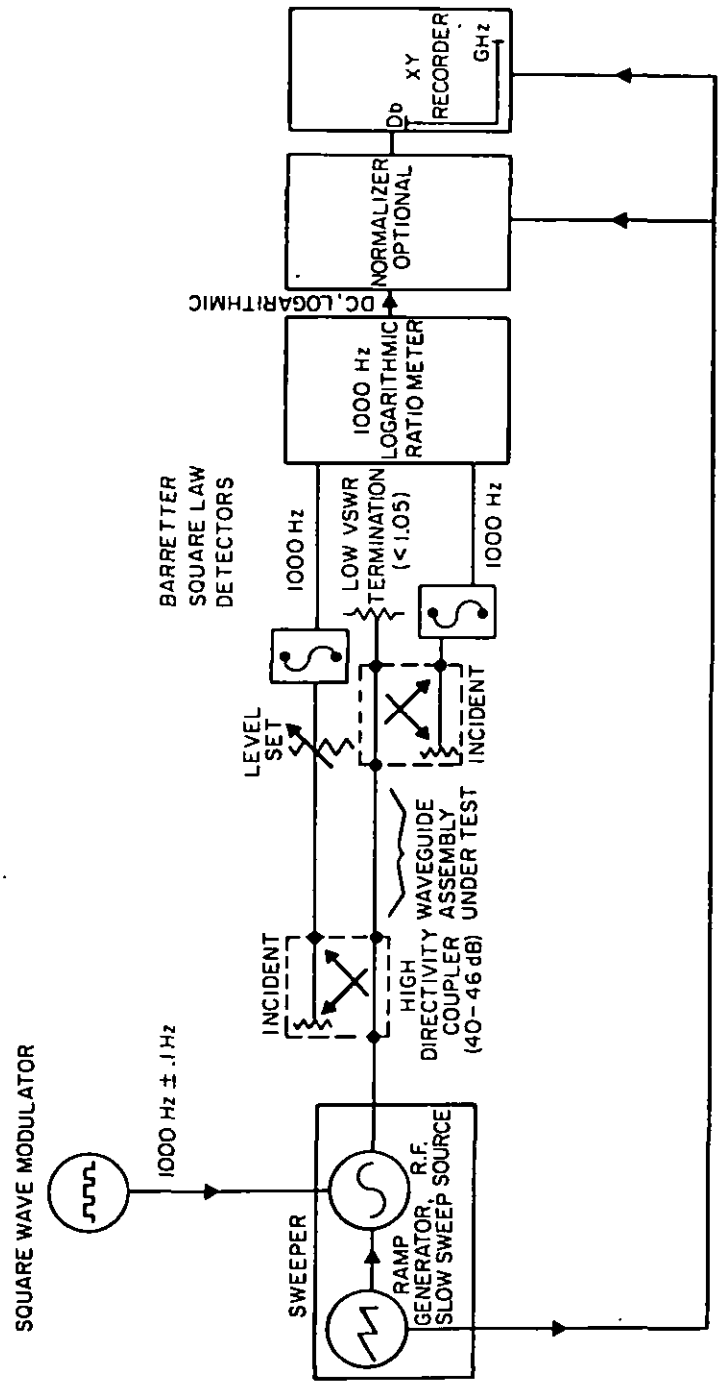



FIGURE 2. Attenuation test setup.



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b. Polarization voltage - Not applicable.
c. Loading voltage - Not applicable.
d. Measurements after test - VSWR and attenuation shall be measured as specified in 4.7.2 and 4.7.3, respectively.

4.7.10 Thermal shock (see 3.13). Waveguides shall be tested in accordance with MIL-STD-202, method 107. The following details shall apply:

a. Special mounting - Not applicable.
b. Test condition letter - A.
c. Measurements before test - Not applicable.
d. Measurements after test - VSWR and attenuation shall be measured as specified in 4.7.2 and 4.7.3, respectively.

4.7.11 Power handling capability (when specified, see 3.1 and 6.2) (see 3.14). The waveguide shall be subjected to the radio frequency power level specified (see 3.1) at the frequency, simulated altitude, and temperature specified (see 3.1). Power shall be maintained for 1 hour after the temperature of the waveguide has reached equilibrium (temperature shall not vary more than 5°C within a 15-minute period). During the test, the waveguide shall be terminated in a matched load (1.05 VSWR, maximum). After the test, the VSWR shall be measured as specified in 4.7.2.

4.7.12 Surface roughness (see 3.15). The average interior surface roughness of the waveguide shall be measured using a brush type profilometer. A 1-foot length shall be cut in half longitudinally. Then both lateral and longitudinal surface roughness tests shall be made over the entire length of the two halves.

5. PACKAGING

5.1 Preservation-packaging . Preservation-packaging shall be level A or C, or as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Waveguides shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Waveguides shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging.

5.1.1.4.1 Waveguides 3 feet or less in length. Unless otherwise specified (see 6.2), waveguides 3 feet or less in length shall be individually packaged in accordance with MIL-P-116, submethod IA-8 insuring compliance with the applicable requirements of that specification. Each unit package shall be placed in a supplementary container conforming to PPP-B-566, PPP-B-676, PPP-T-495 or PPP-B-636 except that waveguides exceeding 1 foot in length shall be restricted to supplementary containers conforming to PPP-B-636.

5.1.1.4.2 Waveguides exceeding 3 feet in length. Unless otherwise specified (see 6.2), waveguides exceeding 3 feet in length shall be individually packaged in accordance with MIL-P-116, method IA as follows: The open ends of each waveguide shall be covered by a water-vaporproof barrier material conforming to MIL-B-131, class 1 or MIL-F-22191, type I. This barrier material shall extend a minimum of six inches back from each open end and shall be secured and sealed with tape conforming to PPP-T-66, type I. When the opening exceeds the width of the tape, cross strips shall be used to reinforce the barrier material. Alternately, the opening may be sealed with cover plates or caps and plugs secured with tape conforming to PPP-T-66, type I. All sealing devices shall meet the water-vaporproof requirements of MIL-B-131. The unit container for flexible waveguides exceeding 3 feet in length and for semirigid waveguides exceeding 20 feet in length shall conform to the requirements for coils or reels specified in MIL-C-12000. The unit container for semirigid waveguides not exceeding 20 feet in length shall conform to the level of packing specified (see 5.2).



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5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. Waveguides shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C, or as specified (see 6.2). The packing for flexible waveguides exceeding 3 feet in length and for semirigid waveguides exceeding 20 feet in length shall conform to the packing requirements of MIL-C-12000. The packing for flexible waveguides not exceeding 3 feet in length and for semirigid waveguides not exceeding 20 feet in length shall be as specified herein.

5.2.1 Level A.

5.2.1.1 Flexible or semirigid waveguides 3 feet or less in length. These packaged waveguides shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joints with tape, 2 inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix of PPP-B-636 using nonmetallic or tape banding only.

5.2.1.2 Semirigid waveguides over 3 feet but not exceeding 6 feet in length. These packaged waveguides shall be individually packed in a fiberboard container conforming to PPP-B-636, class weather resistant, style optional, special requirements. Blocking and bracing shall be provided as necessary in accordance with MIL-STD-1186 to prevent movement of the waveguide within the container. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joint with tape, 2 inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements shall be applied in accordance with the appendix of PPP-B-636 using nonmetallic or tape banding only.

5.2.1.3 Semirigid waveguides over 6 feet but not exceeding 20 feet in length. These packaged waveguides shall be individually packed in a wood box conforming to PPP-B-601, overseas type, grade A or PPP-B-621, class 2, grade A. Blocking and bearing shall be provided as necessary in accordance with MIL-STD-1186 to prevent movement of the waveguide within the container closure and strapping requirements shall be as specified in the applicable specification.

5.2.2 Level B. Packing shall be as specified in 5.2.1.1, 5.2.1.2 and 5.2.1.3 except that the container shall conform to the domestic type or class of the respective specification.

5.2.3 Level C. The packaged waveguides shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

5.2.4 Unitized loads. Unless otherwise specified (see 6.2), unitized loads, commensurate with the level of packing specified in the contract shall be used at the option of the contractor to consolidate packed waveguides for ease in handling and shipment. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.



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5.3 Marking. In addition to any special marking required by the contract (see 6.2), each unit package, supplementary and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.6.2.

5.4.3 Army procurements.

5.4.3.1 Level A (maximum military protection) unit packaging. MIL-P-116 submethod IC-1 and method IC shall be used in lieu of submethod IA-8 and method IA, respectively. All supplementary containers shall either be weather resistant or overwrapped with waterproof barrier materials (see 5.1.1.4.1, 5.1.1.4.2, and 6.2).

5.4.3.2 Level A (maximum military protection) and level B (minimum military protection) packing. The fiberboard containers specified under level A packing shall not be banded but shall be placed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2 style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specification except that metal strapping shall conform to QQ-S-781, type I finish A. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. If not described in the container specification, the skids shall be applied in a manner which will adequately support the item and facilitate the use of material handling equipment. For level B packing, fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded (see 5.2.1.1, 5.2.1.2, and 5.2.2).

5.4.3.3 Commercial packaging. Commercial packaging (including unit packaging, packing and marking) shall be in accordance with Federal Standard No. 356.

6. NOTES

6.1 Intended use. Waveguides covered by this specification are intended for use as radio frequency transmission lines in military services electronic equipment.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. The complete part number (see 1.2 and 3.1).
c. Inspection of commercial packaging (see 4.6.2).
d. Levels of preservation-packaging and packing required (see 5.1 and 5.2).
e. Method of preservation, if other than submethod IA-8 or method IA (see 5.1.1.4).
f. Special marking, if required (see 5.3).
g. Requirements for power handling capability testing (see 3.1 and 3.14).
h. When a first article is required for inspection and approval (see 3.2, 4.5, and 6.3).
i. If unitized loads are required (and not at the option of the contractor) (see 5.2.4).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a preproduction sample, a first production item or it may be a standard production item from the contractors current



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inventory. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army -
Navy - SH
Air Force -

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5985-0861)

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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